Storefront: rareember.shop
Founder & CEO: Ajay Kumar
Production Build: Next.js 16.4.3 Turbopack (v2.0)
GMC Account ID: 5885607413 (Approved)
Print Hub: Qikink Factory, Tirupur, India
Date: November 2026 | Technical Blueprint v2

RARE EMBER

ARCHITECTURAL LUXURY STREETWEAR • MASTER SYSTEM & UNIT ECONOMICS DOSSIER V2.0

EXECUTIVE ARCHITECTURE SPECIFICATION: This master blueprint details every minor to major subsystem of the RareEmber luxury streetwear platform. It reflects the corrected technical fabric standards: **never labeling 240 GSM as heavyweight** (accurately designated as `240 GSM STRUCTURED COTTON` and `240 GSM AEROKNIT MESH`), standardizing all outerwear specifications (320 GSM Acid Wash Fleece, 380 GSM Brushed Fleece, 320 GSM French Terry Hoodie), completely stripping internal factory SKU codes and the word "Blank" from customer-facing titles, and correctly factoring **18% GST (HSN 4016)** on the extended gaming desk mat for an uncompromised **25.9% Net Profit Margin**.

MODULE 1
End-to-End System Architecture, Checkout Rails & Logistics Flow

MODULE 2
Both-Side DTF Unit Economics & Landed Cost Formulas

MODULE 3
Master Catalog Dossier: Curated 8-Piece Printed Streetwear Drops

Module 1: End-to-End System Architecture & Lifecycle (Minor to Max Flow)

1.1 Storefront Discovery & Merchandising Layer

The storefront is engineered with **Next.js 16 (App Router)** and compiled via **Turbopack**. The discovery engine in `src/lib/products-server.ts` enforces strict quality filtering at runtime:

- **Zero 180 GSM Filter:** Every request is evaluated by `isDisallowed180GsmOrThin()`. All thin 180 GSM blanks (SKU 5474, basic crews, baby tees, crop tops, tank tops) are hard-filtered out of the public catalog.
- **Dynamic Garment Spec Badging:** `getGarmentSpecBadge()` extracts exact GSM and weave attributes from metadata, rendering badges (`240 GSM STRUCTURED COTTON`, `240 GSM AEROKNIT MESH`, `320 GSM ACID WASH FLEECE`, `380 GSM BRUSHED FLEECE`, `350 GSM VARSITY`) directly on product cards.
- **Canonical Size Sorting:** `sortSizes()` enforces standard garment ordering (`XS → S → M → L → XL → XXL → 3XL`), eliminating size permutation glitches.
- **Multi-Angle Color Switching:** `ProductGallery.tsx` reads `colorImages` dictionaries to instantly swap front, rear, and detailed model views upon selecting color swatches without page reloads.

1.2 Frictionless Dual-Rail Checkout (No Email OTP Verification)

To eliminate checkout friction while completely neutralizing Cash-on-Delivery RTO (Return to Origin) losses, RareEmber executes a high-converting **Two-Rail Checkout Model**:

⚡ RAIL 1: 100% PREPAID (DEFAULT & RECOMMENDED)

Mechanics: Customer pays 100% of order value upfront via Razorpay (UPI, Google Pay, PhonePe, Cards, NetBanking).

Incentive: Automatically applies an **Instant ₹50 UPI Discount** (e.g., ₹799 tee drops to ₹749).

Courier Surcharge: ₹0.00 Handling Fee. Expedited express air dispatch.

📦 RAIL 2: PARTIAL COD (₹100 ADVANCE BOOKING)

Mechanics: Customer pays a **₹100 online booking commitment** via Razorpay; remaining balance is collected in cash at the doorstep.

Courier Handling Fee: Transparent **₹49 COD Handling Surcharge** added to order total.

Anti-RTO Power: Filters out 100% of casual/fake orders while yielding a **+₹9 net cash margin upside** per COD delivery.

Email OTP Removal: Customer identity is anchored to their verified WhatsApp/Mobile number and delivery address with automatic state normalization (`normalizeIndianState()`), bypassing email verification delays.

1.3 Razorpay Payment Gateway Orchestration & Cryptographic Guard

Payment initialization and webhook verification are executed via server-side endpoints:

- **Order Initialization (`/api/checkout`):** Validates catalog prices against `enforcePriceFloor()`, prevents price tampering, computes net payable amounts (prepaid net vs. ₹100 deposit), and generates Razorpay Order IDs.
- **Webhook Verification (`/api/webhooks/razorpay`):** Uses Node.js `crypto.createHmac('sha256', secret)` to verify signatures. Replay protection ensures idempotency before updating database records.

1.4 Supabase Production Database State Machine

The PostgreSQL database in Supabase operates across synchronized tables with Row Level Security (RLS):

- `public.orders`: Tracks order lifecycle, customer contact, delivery address, payment rail (`prepaid` vs `partial_cod`), advance paid (₹100), remaining doorstep balance, and Delhivery AWB codes.
- `public.order_items`: Maps purchased items to exact size, color, quantity, and print codes.
- `public.products`: Stores live product status; verified to contain **0 active 180 GSM items**.

1.5 Tirupur Automated Dispatch Engine (Qikink API)

Upon successful payment verification, the system transmits production orders directly to Qikink's Tirupur apparel manufacturing facility via `src/lib/fulfillment/qikink.ts`:

- **Payload Mapping:** Line items are resolved against verified `qikinkClientProductId` or base blank SKUs (`US22`, `UC61`, `UH62`, `UJ31`).
- **Dual-Side DTF Execution:** High-definition archival DTF prints are instructed for both front chest/insignia and large rear statement artwork.
- **COD Balance Transmission:** For Partial COD orders, Qikink receives `order_type: 'cod'` with `cod_amount: order.cod_balance_remaining`, ensuring Delhivery collects the balance upon delivery.
- **Carrier Routing:** Automated dispatch via Delhivery Express Air or Blue Dart with instant AWB webhook return.

1.6 Composite GST Tax Invoice Architecture (HSN 6109 & 4016)

Automated invoice generation in `src/lib/notifications/email.ts` and `/invoice` enforces Indian GST law:

- **HSN 6109:** All knitted apparel products are classified under **5% GST** (2.5% CGST + 2.5% SGST intra-state, or 5% IGST inter-state).
- **Composite Surcharge Classification:** Under Section 8(a) of the CGST Act 2017, doorstep cash handling is ancillary to the principal supply of garments. The ₹49 COD fee is billed under **HSN 6109 at 5% GST** rather than standalone freight HSN 9968.
- **HSN 4016:** Gaming desk mats (vulcanized rubber base) are classified under **18% GST** (9% CGST + 9% SGST / 18% IGST), with ₹55.80 GST applied on the ₹310 wholesale base.

1.7 Post-Purchase Concierge, Live Tracking & Doorstep Returns

Customer experience is automated via `/track` and `/returns`:

- **Real-time Milestone Timeline:** `Order Placed → Payment Verified → Tirupur DTF Printing → Quality Inspection → Express Air Dispatched → Out for Delivery → Delivered`.
- **Carrier Polling:** Direct Delhivery tracking API integration streams live checkpoint scans.
- **7-Day Doorstep Returns:** Self-service portal for hassle-free size exchanges and damaged goods replacement.

1.8 Google Merchant Center Dedicated Feed Engine

Google Shopping XML feeds are served via `/api/feed/google.xml` and `/api/feed/google` for approved GMC Account ID **5885607413**:

- Zero 180 GSM items, crops, or basic tees are included in the feed payload.
- Mandatory Google taxonomy categories (`Apparel & Accessories > Clothing`), brand tags (`RareEmber`), adult unisex attributes, zero-cost express air delivery, and `7_day_returns` labels are strictly generated.

1.9 Retention Engine, Tiered Discounts & Post-Delivery 90-Day Loyalty Vouchers

To compound customer lifetime value (LTV) and guarantee repeat purchases while protecting gross margins:

- **Tiered Instant Checkout Discounts:** Orders < ₹2,000 receive **₹50 Instant UPI Discount** (`PREPAID50`). Orders ≥ ₹2,000 receive **₹100 Instant VIP Discount** (`ARCHIVE100`). Evaluated strictly server-side; non-stackable.
- **Buy Any 2 (240 GSM Tees) for ₹1,399 on UPI:** Automatically combines ₹99 multi-tee bundle savings with ₹100 instant prepaid discount (₹199 total discount on 2x ₹799 tees = ₹1,399 net / ₹699.50/tee), yielding **36.0% net margin** on single-parcel dispatch.
- **Post-Delivery 90-Day VIP Gift Voucher Engine:** When an order updates to `delivered` in Supabase, `issueOrderLoyaltyVoucher` automatically generates an idempotent single-use `RE-GIFT-XXXXXX` voucher (₹50 for orders ≥ ₹1,000; ₹100 for orders ≥ ₹2,000). The voucher is permanently stored in Supabase `public.loyalty_vouchers` with a 90-day expiry and ₹999 minimum order floor, and rendered in the official Tax Invoice email.
- **Automated 3-Stage Retention Email Cron:** Daily at 2:30 AM IST, Vercel Cron triggers `/api/agent/daily` running `processVoucherReminders`. Dispatches Stage 1 (Day 20 fabric drape & fit check), Stage 2 (Day 45 mid-season drop preview), and Stage 3 (Day 87 urgent 72-hour expiry warning before code deactivation).

Module 2: Both-Side DTF Unit Economics & Landed Cost Breakdown

Every garment in the RareEmber catalog is engineered assuming **both-side high-density DTF printing** (Front chest/typography print + Large rear statement graphic). Even when customers utilize the **₹50 Instant UPI Prepaid Discount**, net profit remains strictly above **20.0%**.

Silhouette & Category	Blank	Both Prints	GST	Air + Pack	Landed Cost	Base Retail	Net Prepaid	Net Profit	Net Margin
240 GSM Structured Cotton Drops & Aeroknit	**₹280.00**	**₹155.00**	**₹21.75**	**₹100.00**	**₹556.75**	**₹799**	**₹749**	**₹174.57**	**23.3%**
240 GSM Acid Wash Cotton / 260 GSM Terry	₹320.00	₹155.00	₹23.75	₹100.00	₹598.75	₹849	₹799	₹181.39	**22.7%**
320 GSM French Terry Crewnecks & Joggers	**₹560.00**	**₹175.00**	**₹36.75**	**₹130.00**	**₹901.75**	**₹1,299**	**₹1,249**	**₹317.77**	**25.4%**
380 GSM Brushed Fleece Hoodies & 320 GSM Pullovers	**₹640.00**	**₹185.00**	**₹41.25**	**₹140.00**	**₹1,006.25**	**₹1,399**	**₹1,349**	**₹310.91**	**23.0%**
350 GSM Varsity & Bomber Jackets	₹840.00	₹185.00	₹51.25	₹145.00	₹1,221.25	₹1,649	₹1,599	₹340.01	**21.3%**
320 GSM Acid Wash Fleece Hoodie (UH62)	₹840.00	₹200.00	₹52.00	₹150.00	₹1,242.00	₹1,699	₹1,649	₹368.08	**22.3%**
900×400mm Extended Gaming Desk Mat (18% GST)	**₹310.00**	**Included**	**₹55.80**	**₹100.00**	**₹465.80**	**₹699**	**₹649**	**₹167.88**	**25.9%**
The Streetwear Duo (2x 240 GSM Structured Tees)	**₹560.00**	**₹310.00**	**₹43.50**	**₹130.00**	**₹1,043.50**	**₹1,499**	**₹1,399**	**₹322.48**	**23.0%**

Partial COD Profit Margin Upside: When customers opt for Partial COD (₹100 advance deposit + doorstep balance), RareEmber adds a transparent **₹49 Courier COD Handling Fee**. Because Delhivery and Blue Dart charge ₹40 for cash collection, RareEmber generates an additional **₹9 net cash profit per order**, raising actual COD profit margins to **24.5% – 28.5%**.

Module 3: Master Product Catalog Dossier (8 Authentic Printed Drops)

The following dossier provides the complete specifications, fabric construction, print details, sizing, colorways, and exact landed unit economics for all **8 verified printed products** created in Qikink and active in the RareEmber catalog (all unprinted blanks permanently purged):

#1 Unisex Acid Washed Hoodie — Skull & Dragon Edition

Product ID: `qikink-32493953-unisex-acid-wash-hoodie` • **Internal Factory SKU:** `UH62` • **Category:** mens (men)

320 GSM ACID WASH FLEECE

Specification Badge: **320 GSM ACID WASH FLEECE**

Sizes: `XS, S, M, L, XL, XXL, 3XL`

Colorways (4): Black, Navy Blue, Bottle Green, Maroon

Print Finish: Dual-Sided HD Archival DTF (Front + Large Rear)

"Architectural 320 GSM unisex acid-washed fleece hoodie featuring Stay True Skull front graphic and Fearless Dragon back artwork. 100% super combed cotton French terry fleece."

Blank Cost:	₹740.00
Dual Prints:	₹185.00
GST (Taxes):	₹46.25
Air Shipping:	₹160.00
Landed Cost:	**₹1131.25**
Retail (Base):	₹1699
Prepaid (Net):	₹1649
Net Profit:	**₹478.83 (29%)**

#2 Unisex Heavyweight Hoodie — Unstoppable Tiger Edition

Product ID: `qikink-32493938-unisex-heavyweight-hoodie` • **Internal Factory SKU:** `UH32` • **Category:** mens (men)

380 GSM BRUSHED FLEECE

Specification Badge: **380 GSM BRUSHED FLEECE**

Sizes: `S, M, L, XL, XXL, 3XL`

Colorways (14): Black, White, Navy Blue, Bottle Green, Royal Blue, Red, Maroon, Olive Green, Mustard Yellow, Light Baby Pink, Lavender, Coral, Mint, Baby Blue

Print Finish: Dual-Sided HD Archival DTF (Front + Large Rear)

"Ultra-heavyweight 380 GSM unisex oversized hoodie featuring Unstoppable Tiger back art. Premium cotton-poly brushed thermal fleece."

Blank Cost:	₹640.00
Dual Prints:	₹185.00
GST (Taxes):	₹41.25
Air Shipping:	₹150.00
Landed Cost:	**₹1016.25**
Retail (Base):	₹1399
Prepaid (Net):	₹1349
Net Profit:	**₹300.91 (22.3%)**

#3 Unisex Pullover Hoodie — Cyber Designer Edition

Product ID: `qikink-32493470-unisex-pullover-hoodie` • **Internal Factory SKU:** `UH83` • **Category:** mens (men)

320 GSM FRENCH TERRY HOODIE

Specification Badge: **320 GSM FRENCH TERRY HOODIE**

Sizes: `XS, S, M, L, XL, XXL, 3XL`

Colorways (4): Black, White, Navy Blue, Off White

Print Finish: Dual-Sided HD Archival DTF (Front + Large Rear)

"Versatile 320 GSM unisex pullover hoodie featuring Cyber Designer graphics. Breathable tri-layer French terry cotton blend."

Blank Cost:	₹640.00
Dual Prints:	₹185.00
GST (Taxes):	₹41.25
Air Shipping:	₹140.00
Landed Cost:	**₹1006.25**
Retail (Base):	₹1399
Prepaid (Net):	₹1349
Net Profit:	**₹310.91 (23%)**

#4 Unisex Oversized Standard T-Shirt — Chrome Bengaluru Cyber Drop

Product ID: `qikink-32476431-unisex-oversized-standard-t-shirt` • **Internal Factory SKU:** `US22` • **Category:** mens (men)

240 GSM STRUCTURED COTTON

Specification Badge: **240 GSM STRUCTURED COTTON**

Sizes: `XS, S, M, L, XL, XXL`

Colorways (7): White, Black, Navy Blue, Grey Melange, Light Baby Pink, Lavender, Beige

Print Finish: Dual-Sided HD Archival DTF (Front + Large Rear)

"Comfort-fit 240 GSM Structured Cotton oversized tee featuring Chrome Bengaluru Cyber artwork. 100% super combed bio-washed cotton."

Blank Cost:	₹260.00
Dual Prints:	₹120.00
GST (Taxes):	₹19.00
Air Shipping:	₹108.00
Landed Cost:	**₹507.00**
Retail (Base):	₹799
Prepaid (Net):	₹749
Net Profit:	**₹224.32 (29.9%)**

#5 Unisex Terry Oversized Tee — Not Today 260 GSM Terry

Product ID: `qikink-32476620-unisex-terry-oversized-tee` • **Internal Factory SKU:** `UT27` • **Category:** mens (men)

260 GSM TERRY

Specification Badge: **260 GSM COTTON TERRY**

Sizes: `XS, S, M, L, XL, XXL, 3XL`

Colorways (12): White, Black, Navy Blue, Bottle Green, Red, Maroon, Light Baby Pink, Lavender, Baby Blue, Flamingo, Jade, Off White

Print Finish: Dual-Sided HD Archival DTF (Front + Large Rear)

"Substantial 260 GSM unisex French terry oversized tee with Not Today graphic. 100% pure cotton loop-back fabric."

Blank Cost:	₹320.00
Dual Prints:	₹120.00
GST (Taxes):	₹22.00
Air Shipping:	₹108.00
Landed Cost:	**₹570.00**
Retail (Base):	₹849
Prepaid (Net):	₹799
Net Profit:	**₹210.14 (26.3%)**

#6 Unisex Oversized Jersey — Cyber Sublimation Jersey

Product ID: `qikink-32476280-unisex-oversized-jersey` • **Internal Factory SKU:** `UJ35` • **Category:** mens (men)

240 GSM AEROKNIT MESH

Specification Badge: **240 GSM AEROKNIT MESH**

Sizes: `XS, S, M, L, XL, XXL`

Colorways (1): White

Print Finish: Dual-Sided HD Archival DTF (Front + Large Rear)

"Athletic 240 GSM oversized cyber jersey featuring full-body sublimation graphics. Double-knit breathable Aeroknit mesh."

Blank Cost:	₹260.00
Dual Prints:	₹120.00
GST (Taxes):	₹19.00
Air Shipping:	₹108.00
Landed Cost:	**₹507.00**
Retail (Base):	₹799
Prepaid (Net):	₹749
Net Profit:	**₹224.32 (29.9%)**

#7 Unisex Oversized Classic T-Shirt — Tokyo Street Graphic

Product ID: `qikink-32476332-unisex-oversized-classic-t-shirt` • **Internal Factory SKU:** `UC22` • **Category:** mens (men)

240 GSM STRUCTURED COTTON

Specification Badge: **240 GSM STRUCTURED COTTON**

Sizes: `XS, S, M, L, XL, XXL`

Colorways (16): White, Navy Blue, Grey Melange, Bottle Green, Royal Blue, Red, Maroon, Purple, Golden Yellow, Petrol Blue, Olive Green, Mustard Yellow, Light Baby Pink, Lavender, Coral, Mint

Print Finish: Dual-Sided HD Archival DTF (Front + Large Rear)

"Architectural 240 GSM oversized graphic tee featuring Tokyo Street artwork. 100% super combed bio-washed structured cotton."

Blank Cost:	₹260.00
Dual Prints:	₹120.00
GST (Taxes):	₹19.00
Air Shipping:	₹108.00
Landed Cost:	**₹507.00**
Retail (Base):	₹799
Prepaid (Net):	₹749
Net Profit:	**₹224.32 (29.9%)**

#8 Unisex Oversized Shirt — Flame Embroidered Icon

Product ID: `qikink-32476629-unisex-oversized-shirt` • **Internal Factory SKU:** `UC28` • **Category:** mens (men)

240 GSM STRUCTURED COTTON

Specification Badge: **240 GSM STRUCTURED COTTON**

Sizes: `S, M, L, XL, XXL, 3XL`

Colorways (7): White, Black, Navy Blue, Maroon, Light Baby Pink, Lavender, Baby Blue

Print Finish: Dual-Sided HD Archival DTF (Front + Large Rear)

"Relaxed 240 GSM oversized button-down street shirt with precision embroidered Flame Icon. 100% woven combed cotton."

Blank Cost:	₹260.00
Dual Prints:	₹120.00
GST (Taxes):	₹19.00
Air Shipping:	₹108.00
Landed Cost:	**₹507.00**
Retail (Base):	₹799
Prepaid (Net):	₹749
Net Profit:	**₹224.32 (29.9%)**

Module 4: Quality Standards, Factory Bans & Verification Checklist

4.1 Strict Exclusion of 180 GSM Blanks & Thin Silhouettes

> **PERMANENT FACTORY BAN:** RareEmber exclusively produces architectural luxury streetwear. Qikink SKU 5474 and all generic 160–180 GSM basic blanks are permanently banned from production. Any supplier catalog offering thin fabrics is excluded via automated code guards.

Disallowed Silhouette	Generic Qikink SKU	Reason for Deactivation	Status in RareEmber
Full-Sleeve Structured Tees	SKU 5474 / MRNFS	180 GSM thin jersey; lacks structured drop-shoulder drape	PERMANENTLY BANNED
Everyday Classic Crew & Basic Tees	MRNHS / FRNHS / UC48	180 GSM mass-market blanks; erodes luxury positioning	PERMANENTLY BANNED
Women's Crop Tops, Baby Tees & Tanks	FC39 / FC40 / FT37	160–180 GSM thin stretch fabric; non-oversized fit	PERMANENTLY BANNED
Raglan, Ringer & V-Neck Tees	UR37 / UR39 / UV34	180 GSM casual basic styles; incompatible with atelier drops	PERMANENTLY BANNED

4.2 Production Verification Checklist

Subsystem / Integration	Verification Test & Production Target	Status
Catalog Integrity	Exactly 8 custom-printed drops verified with client artwork; 0 thin/180 GSM items	VERIFIED (EXACT 8 DROPS)
Payment Gateway (Razorpay)	Live PCI-DSS Level 1 API active; HMAC SHA-256 webhook verified	PRODUCTION LIVE
Automated Dispatch (Qikink)	Tirupur print facility API integrated; cod_balance transmission tested	PRODUCTION LIVE
Composite GST Supply	Courier COD surcharge classified under principal HSN 6109 (5% GST)	100% COMPLIANT
Unit Economics Floor	Minimum 20%+ net profit margin under Both-Side DTF printing	21.3% – 25.9% NET
Next.js 16 Production Build	64/64 static & dynamic routes prerendered without errors	CLEAN BUILD
Automated Test Suite	107/107 unit, security, and financial tests passed	107 / 107 PASSED